COLLEGE OF COMPUTING AND INFORMATICS

PUTRAJAYA CAMPUS

ASSIGNMENT PART 3

SEMESTER 2 2025/2026

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SUBJECT : SOFTWARE CONSTRUCTION & METHODS

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NUMBER OF GROUP : GROUP 4

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1.0 INTRODUCTION

1.1 PROJECT BACKGROUND

With the rapid growth of digital technology in education, learning institutions are increasingly relying on online platforms to manage teaching and learning activities. Traditional classroom-based management methods such as manual assignment submission, physical attendance tracking, and paper-based grading are inefficient and difficult to scale.

A Student Learning Management System (SLMS) is a web-based application designed to support digital learning processes. Similar to established platforms like Moodle, an SLMS enables students, lecturers, and administrators to interact within a centralized system. It allows course management, assignment submission, grading, material distribution, and discussion forums to be handled electronically.

The proposed SLMS will serve as a simplified academic platform that manages users, courses, learning materials, assignments, submissions, and discussions. The system emphasizes proper software construction practices, clean architecture, and structured development to ensure maintainability and scalability.

This project is developed as part of the Software Construction & Maintenance course to apply theoretical knowledge into practical system design and implementation.

1.2 PROJECT OBJECTIVE

The main objective of this project is to design and develop a Student Learning Management System (SLMS) by applying proper software construction principles and structured programming practices. A Student Learning Management System is a software platform used in educational institutions to manage academic information such as courses, students, and their relationships. The system helps organize course information, maintain student records, and support academic management activities in a structured and efficient way.

In this project, the SLMS focuses on managing course profiles and student profiles, allowing users to perform operations such as creating, searching, editing, and deleting records. The system also manages the relationship between students and courses, enabling the tracking of course enrolments. The development process follows modular design, object-oriented programming principles, and proper coding standards to ensure the system is scalable, readable, and maintainable.

The specific objectives of this project are:

1. To design the overall structure of the Student Learning Management System using use case diagrams and class diagrams.
2. To develop modules that allow manual input and management of course profiles, including course name, course code, credit hour, course summary, and MS Teams link.
3. To implement a student profile management module that stores and manages student information such as student ID, name, email, and phone number.
4. To implement system functionalities such as searching, editing, deleting, and displaying records for both courses and students.
5. To manage the relationship between students and courses, allowing students to be assigned to courses and courses to maintain a list of enrolled students.
6. To apply software construction best practices, including modular programming, proper naming conventions, documentation standards, and coding standards.
7. To implement error handling and input validation to ensure the system handles invalid inputs and potential errors effectively.

8. To demonstrate the use of version control and continuous integration through Github collaboration during the development process.

Through this project, students will demonstrate their understanding of structured software planning, object-oriented design, modular programming, and disciplined software development practices in building a simplified academic management system similar to existing learning platforms such as Moodle.

2.0 USE CASE AND CLASS DESIGN OF SLMS

2.1 USE CASE DIAGRAM

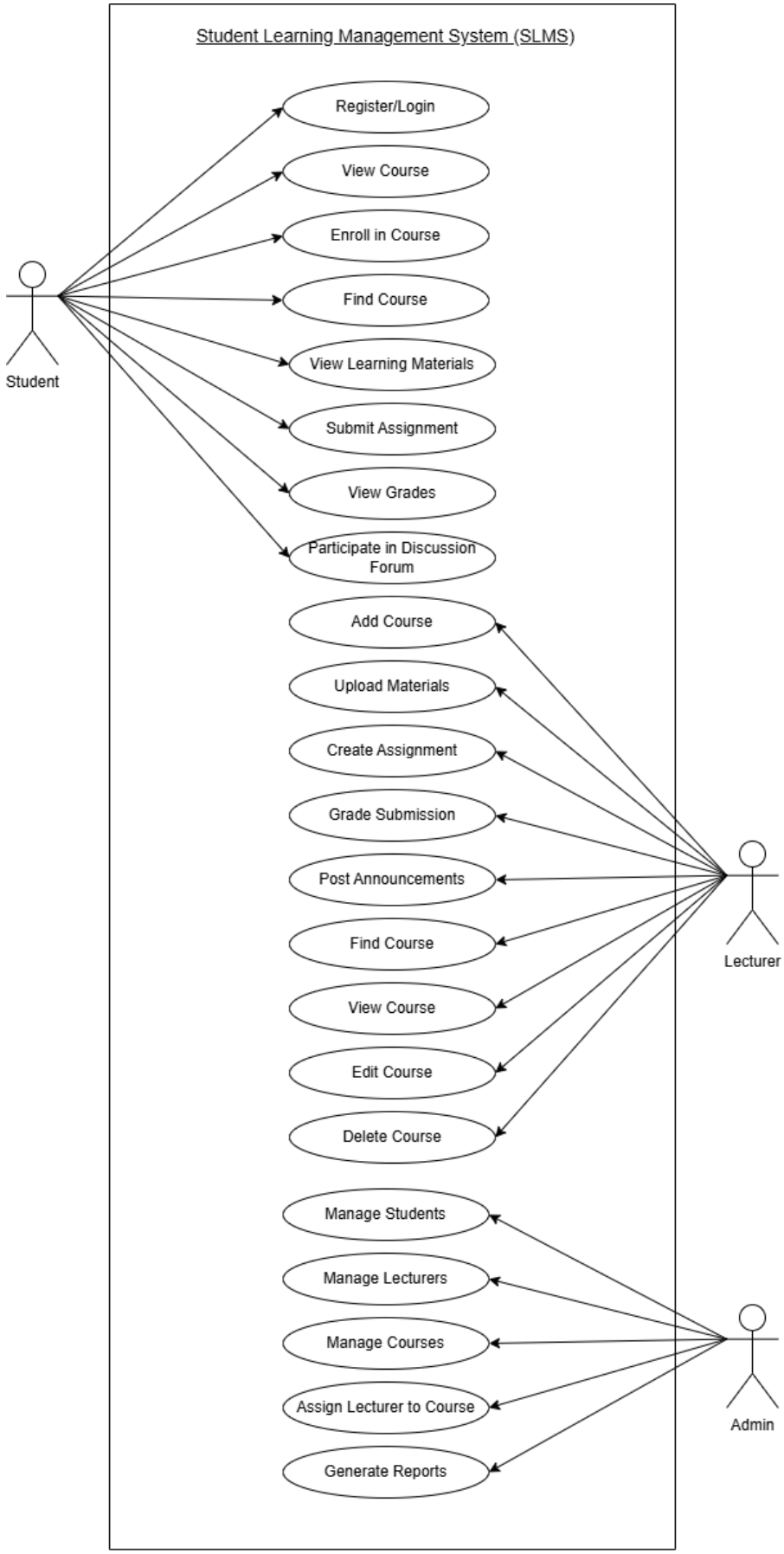
2.1.1 USE CASE DIAGRAM DESCRIPTION

A use case describes how users interact with a system to achieve a specific goal. It represents a sequence of actions between an actor and the system that leads to a useful outcome.

Use cases help developers understand:

- System functionality
- User requirements
- System interactions
- Scope of the system

Use case diagrams provide a high-level overview of the system's functionality and how external actors interact with it.


Student Learning Management System (SLMS)
Register/Login
View Course
Enroll in Course
Find Course
View Learning Materials
Submit Assignment
View Grades
Participate in Discussion Forum
Add Course
Upload Materials
Create Assignment
Grade Submission
Post Announcements
Find Course
View Course
Edit Course
Delete Course
Manage Students
Manage Lecturers
Manage Courses
Assign Lecturer to Course
Generate Reports
Student
Lecturer
Admin

2.1.2 ACTORS DESCRIPTION

2.1.2.1 STUDENT

A Student is a user who participates in courses and learning activities. Students can access course materials, submit assignments, participate in discussions, and view grades.

Student responsibilities include:

- Register and log into the system
- Enrol in available courses
- View course
- View learning materials
- Find course
- Submit assignments
- Participate in discussion forums
- View grades

2.1.2.1 LECTURER

A Lecturer is responsible for managing courses and delivering learning content to students.

Lecturer responsibilities include:

- Add courses
- Upload learning materials
- Create assignments
- Grade student submissions
- Post announcements
- Find course
- View course
- Edit course
- Delete course

2.1.2.3 ADMIN

An Admin is responsible for managing the system and controlling overall operations.
Admin responsibilities include:

- Manage students
- Manage lecturers
- Manage courses
- Assign lecturers to courses
- Generate system reports

2.1.3 SYSTEM BOUNDARY

The system boundary represents the limits of the SLMS system. It separates the internal system functionalities from external actors.
In the use case diagram:

- All use cases are located inside the SLMS system boundary.
- Actors such as Student, Lecturer, and Admin are outside the boundary because they interact with the system.

This helps define:

- What the system is responsible for
- What actions are performed by users outside the system

2.1.4 MAIN INTERACTIONS IN THE SYSTEM

The SLMS system supports several key interactions between actors and the system.

2.1.4.1 STUDENTS INTERACTION

Students interact with the system to manage their learning activities.

Main interactions include:

- Registering and logging into the system
- Enrolling in courses
- Viewing learning materials
- Submitting assignments
- Participating in discussions
- Viewing grades
- Viewing course
- Finding course

2.1.4.2 LECTURERS INTERACTION

Lecturers interact with the system to manage course content and evaluate students.

Main interactions include:

- Adding courses
- Viewing courses
- Finding courses
- Editing courses
- Deleting courses
- Uploading learning materials
- Creating assignments
- Grading student submissions
- Providing feedback
- Posting announcements

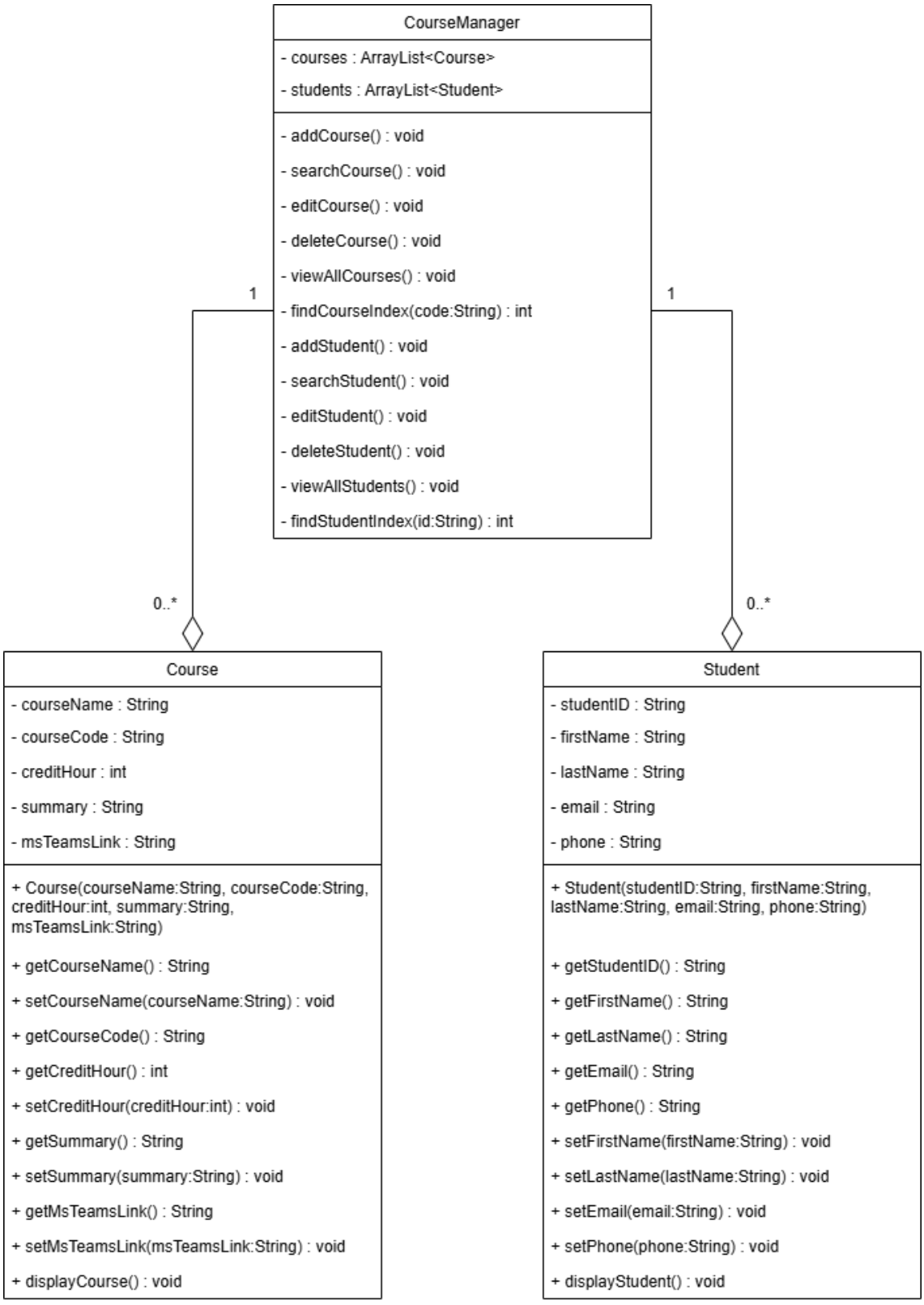
2.1.4.3 ADMIN INTERACTION

Admins interact with the system to maintain and control the platform.

Main interactions include:

- Managing students
- Managing lecturers
- Managing courses
- Assigning lecturers to courses
- Generating system reports

2.2 CLASS DIAGRAM



Object-Oriented Approach

- Used inheritance to promote reusability and reduce redundancy.

Clear Role Separation

Student, Lecturer, and Admin responsibilities are separated to enforce:

- Security
- Maintainability
- Scalability

Proper Relationship Modelling

- Association used for independent entities.
- Aggregation used when object belongs to another but can exist independently.
- Composition used when lifecycle dependency exists.

Extensibility

- Future roles (e.g., TeachingAssistant) can extend User easily.

Maintainability

- Modular class structure makes:
- Debugging easier
- Testing easier
- Feature addition safer

3.0 DOCUMENTATION FORMATTING

The SLMS documentation will follow the following format:

- Font: Times New Roman
- Font Size: 12
- Line spacing: 1.5
- Heading format:
    - 1.0 Main Heading (Bold)
    - 1.1 Subheading
- Alignment: Justified
- UML diagrams labelled properly
- Each section must include:
    - Purpose
    - Description
    - Example (if needed)

For code documentation, use:

- File header comment
- Function header comment
- Inline comments for complex logic

Example:

```
"""
File: course.py
Author: Team SLMS
Description: Handles course-related operations.
Date: 21 Feb 2026
"""
```

4.0 CODING FILE FORMATTING

- Indentation uses 4 spaces.
- Opening and closing brackets must align.
- Comments are required before functions.
- Files organised into folders.

Example:

```
if (loginSuccess) {
    openDashboard();
}
```

5.0 NAMING CONVENTIONS

Element	Convention	Example
Class	PascalCase	StudentAccount
Function	camelCase	submitAssignment()
Variable	snake_case	student_name
Constant	UPPER_CASE	MAX_LOGIN_ATTEMPT
File Name	lowercase	course.py

Term	Meaning
SLMS	Student Learning Management System
LMS	Learning Management System
Enrollment	Student registered in course
Submission	Assignment uploaded by student

6.0 FUNCTION AND METHODS MODULARITY

The system is designed using a modular approach where each function and method performs a single, well-defined task. Large functionalities are broken down into smaller and reusable functions to improve readability, maintainability, and scalability of the system.

For example, user authentication is separated into different functions such as input validation, credential verification, and session handling. This modular design allows easier debugging, testing, and future enhancement of the system.

Each module is designed to minimize dependency on other modules, allowing changes to be made with minimal impact on the overall system. This approach supports team collaboration, as different members can work on different modules independently.

7.0 CONTINUOUS INTEGRATION RULES

The following continuous integration rules are implemented to ensure effective collaboration and maintain code quality throughout the development of the Student Learning Management System (SLMS):

1. Each team member must develop features in a separate feature branch.
2. Direct commits to the main branch are prohibited.
3. All changes must be reviewed and approved by the team leader before merging.
4. Clear and meaningful commit messages are required for every commit.
5. Code must be tested to ensure it functions correctly before integration.
6. GitHub is used as the main platform for version control and collaboration.

These rules help reduce integration issues and improve the overall quality of the system.

Example:
Added assignment submission module.
Fixed login validation error.

8.0 COURSE CLASS

8.1 COURSE CLASS DESIGN

- **Class declaration**:

```
3   public class Course {
```

Explanation:

- Course is the blueprint for a course.
- All course-related data and functions are organized inside this class.
- Every new course you add will be an object of this class.

- **Attributes**:

```
4       // Attributes
5       private String courseName;
        private String courseCode;
7       private int creditHour;
8       private String summary;
9       private String msTeamsLink;
```

Explanation:

- These are the data for each course.
- private means they are hidden from outside, to protect data integrity.
- courseCode cannot be changed after creation - this ensures uniqueness.

- **Constructor**:

```
11      // Constructor
12      public Course(String courseName, String courseCode, int creditHour, String summary, String msTeamsLink) {
13          this.courseName = courseName;
14          this.courseCode = courseCode;
15          this.creditHour = creditHour;
16          this.summary = summary;
17          this.msTeamsLink = msTeamsLink;
18      }
```

Explanation:

- Constructor is used to create a new course object with all required data.

- **Getters and setters**:

```
20      // Getters and Setters
21      public String getCourseName() { return courseName; }
22      public void setCourseName(String courseName) { this.courseName = courseName; }
23
24      public String getCourseCode() { return courseCode; }
25
26      public int getCreditHour() { return creditHour; }
27      public void setCreditHour(int creditHour) { this.creditHour = creditHour; }
28
29      public String getSummary() { return summary; }
30      public void setSummary(String summary) { this.summary = summary; }
31
32      public String getMsTeamsLink() { return msTeamsLink; }
33      public void setMsTeamsLink(String msTeamsLink) { this.msTeamsLink = msTeamsLink; }
```

Explanation:

- Getters → read/access the data.
- Setters → update/change the data (except courseCode).
- This keeps the data safe and controlled, a principle called encapsulation.

- **Display method**:

```
35      // Display all course attributes
36      public void displayCourse() {
37          System.out.println("Course Name      : " + courseName);
38          System.out.println("Course Code      : " + courseCode);
39          System.out.println("Credit Hour      : " + creditHour);
40          System.out.println("Summary          : " + summary);
41          System.out.println("MS Teams Link    : " + msTeamsLink);
42          System.out.println("---------------------------------------");
43      }
44  }
```

Explanation:

- Method to show all course info in a clear format.
- Useful for View All, Search, Edit, and Delete functions.

8.1 ADD COURSE INPUT IMPLEMENTATION

8.1.1 INPUT FIELDS FOR EACH COURSE ATTRIBUTES

```
57      private void addCourse() {
58          System.out.print("Enter Course Name: ");
59          String name = sc.nextLine();
```

```
61          String code;
62          while (true) {
63              System.out.print("Enter Course Code: ");
64              code = sc.nextLine();
65              if (findCourseIndex(code) != -1) {
66                  System.out.println("Course code already exists! Enter a unique code.");
67              } else break;
68          }
```

```
70          int credit;
71          while (true) {
72              System.out.print("Enter Credit Hour: ");
73              if (sc.hasNextInt()) {
74                  credit = sc.nextInt();
75                  sc.nextLine();
76                  if (credit > 0) break;
77                  else System.out.println("Credit hour must be positive!");
78              } else {
79                  System.out.println("Invalid input! Enter a number.");
80                  sc.next();
81              }
82          }
```

```
84          System.out.print("Enter Course Summary: ");
85          String summary = sc.nextLine();
86
87          System.out.print("Enter MS Teams Link: ");
88          String link = sc.nextLine();
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 1
Enter Course Name: SOFTWARE SECURITY
Enter Course Code: CSEB5113
Enter Credit Hour: 3
Enter Course Summary: SECURITY
Enter MS Teams Link: https://teams.microsoft.co
```

Explanation:

- These are the fields where the user enters course details.
- User inputs: course name, course code, credit hours, summary, and MS Teams link.

8.1.2 'SUBMIT' BUTTON TO ADD COURSE PROFILE

```
90              courses.add(new Course(name, code, credit, summary, link));
91              System.out.println("Course added successfully!");
92          }
```

```
Enter Course Name: SOFTWARE SECURITY
Enter Course Code: CSEB5113
Enter Credit Hour: 3
Enter Course Summary: SECURITY
Enter MS Teams Link: https://teams.mi
Course added successfully!
```

Explanation:

- This is equivalent to a "Submit" button.
- It creates a new Course object and adds it to the ArrayList.
- The confirmation message shows that the course has been successfully added.

8.1.3 STORE MULTIPLE COURSES IN ARRAY/LIST

```
8       // Store multiple courses
        private ArrayList<Course> courses = new ArrayList<>();
        private Scanner sc = new Scanner(System.in);
11  /**
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 5

--- All Courses ---
Course Name     : SOFTWARE SECURITY
Course Code     : CSEB5113
Credit Hour     : 3
Summary         : SECURITY
MS Teams Link   : https://teams.microsoft.com/l/t
----------------------------------------
Course Name     : SOFTWARE CONSTRUCTION & METHOD
Course Code     : CSEB5223
Credit Hour     : 3
Summary         : GITHUB AND CODING
MS Teams Link   : https://teams.microsoft.com/l/t
----------------------------------------
```

Explanation:

- ArrayList stores all courses added by the user.
- Can store any number of courses dynamically, no fixed limit.

8.1.4 PREVENT ARRAY OUT-OF-BOUND ERRORS

```
101             int index = findCourseIndex(code);
102
103             if (index != -1) {
104                 System.out.println("Course found:");
105                 courses.get(index).displayCourse();
106             } else {
107                 System.out.println("Course not found.");
108             }
109         }
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 2
Enter Course Code to search: SCM
Course not found.
```

Explanation:

- Before accessing a course from the list, the program checks if the course exists.
- This prevents array out-of-bound errors (trying to access a non-existing index).

8.2 SEARCH COURSE FUNCTION DEVELOPMENT

8.2.1 IMPLEMENT A LINEAR SEARCH FUNCTION

```
205     private int findCourseIndex(String code) {
206         for (int i = 0; i < courses.size(); i++) {
207             if (courses.get(i).getCourseCode().equalsIgnoreCase(code)) {
208                 return i;
209             }
210         }
211         return -1;
212     }
213 }
```

Explanation:

- This method implements a linear search algorithm.
- The program loops through the ArrayList of courses one by one.
- It compares the course code entered by the user with each course stored in the system.
- If a matching course code is found, the index of that course is returned.
- If no match is found, the method returns -1.

8.2.2 INPUT SEARCH COURSE CRITERIA

```
99              System.out.print("Enter Course Code to search: ");
100             String code = sc.nextLine();
101             int index = findCourseIndex(code);
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: CSEB5223
Invalid input! Enter a number: 2
Enter Course Code to search: CSEB5223
```

Explanation:

- The user enters a course code as the search criterion.
- The program then calls the linear search method (findCourseIndex) to locate the course in the list.

8.2.3 DISPLAY SEARCH COURSE RESULT FOR SUCCESFUL SEARCH

```
103             if (index != -1) {
104                 System.out.println("Course found:");
105                 courses.get(index).displayCourse();
```

```
Enter Course Code to search: CSEB5223
Course found:
Course Name     : SCM
Course Code     : CSEB5223
Credit Hour     : 3
Summary         : GITHUB
MS Teams Link   : https://teams.microsoft.com/l/t
---------------------------------------
```

Explanation:

- If the course exists in the system, the program retrieves the course object using its index.
- The method displayCourse() is called to show all course attributes, including:
    - Course Name
    - Course Code
    - Credit Hour
    - Course Summary
    - MS Teams Link

8.2.4 DISPLAY MESSAGE FOR UNSUCCESSFUL SEARCH

```
106         } else {
107             System.out.println("Course not found.");
108         }
109     }
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 2
Enter Course Code to search: CSEB5113
Course not found.
```

Explanation:

- If the entered course code does not exist, the program shows a message "Course not found."
- This ensures the system handles invalid searches properly without crashing.

8.3 EDIT COURSE FUNCTION DEVELOPMENT

8.3.1 IMPLEMENT SEARCH FUNCTION TO FIND THE TARGET COURSE

```
116             System.out.print("Enter Course Code to edit: ");
117             String code = sc.nextLine();
118             int index = findCourseIndex(code);
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 3
Enter Course Code to edit: CSEB5223
```

Explanation:

- The system first asks the user to enter the course code of the course that needs to be edited.
- The program then calls the linear search method findCourseIndex() to locate the course in the ArrayList.
- If the course exists, its index will be returned for further editing.

8.3.2 DISPLAY COMPLETE COURSE ATTRIBUTES (EDITABLE EXCEPT COURSE CODE)

```
121                 Course c = courses.get(index);
122                 System.out.println("Editing Course: " + c.getCourseCode());
123
124                 System.out.print("New Course Name: ");
125                 c.setCourseName(sc.nextLine());
126
127                 int credit;
128                 while (true) {
129                     System.out.print("New Credit Hour: ");
130                     if (sc.hasNextInt()) {
131                         credit = sc.nextInt();
132                         sc.nextLine();
133                         if (credit > 0) break;
134                         else System.out.println("Credit hour must be positive!");
135                     } else {
136                         System.out.println("Invalid input! Enter a number.");
137                         sc.next();
138                     }
139                 }
140                 c.setCreditHour(credit);
141
142                 System.out.print("New Summary: ");
143                 c.setSummary(sc.nextLine());
144
145                 System.out.print("New MS Teams Link: ");
146                 c.setMsTeamsLink(sc.nextLine());
```

```
Editing Course: CSEB5223
New Course Name: SOFTWARE CONSTRUCTION & METHOD
New Credit Hour: 3
New Summary: GITHUB AND CODING
New MS Teams Link: https://teams.microsoft.com/l
```

Explanation:

- Once the course is found, the program retrieves the course object from the ArrayList.
- The user is allowed to edit all course attributes, including:
    - Course Name
    - Credit Hour
    - Course Summary
    - MS Teams Link
- However, the course code cannot be edited because it acts as a unique identifier for each course.

8.3.3 DISPLAY MESSAGE FOR UNSUCCESSFUL SEARCH

```
150             } else {
151                 System.out.println("Course not found.");
152             }
153         }
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 3
Enter Course Code to edit: CSEB5123
Course not found.
```

Explanation:

- If the course code entered by the user does not exist, the system will display "Course not found."
- This ensures the program handles invalid input without causing system errors.

8.3.4 DISPLAY EDITED COURSE TO VALIDATE CHANGES

```
148                 System.out.println("Course updated successfully!");
149                 c.displayCourse();
```

```
Course updated successfully!
Course Name     : SOFTWARE CONSTRUCTION & METHOD
Course Code     : CSEB5223
Credit Hour     : 3
Summary         : GITHUB AND CODING
MS Teams Link   : https://teams.microsoft.com/l/team/19'
-------------------------------------
```

Explanation:

- After editing the course information, the system displays the updated course details.
- This allows the user to verify that the changes were successfully applied.

8.4 DELETE COURSE FUNCTION DEVELOPMENT

8.4.1 IMPLEMENT SEARCH FUNCTION TO FIND TARGET COURSE

```
160             System.out.print("Enter Course Code to delete: ");
161             String code = sc.nextLine();
162             int index = findCourseIndex(code);
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 4
Enter Course Code to delete: CSEB3123
```

Explanation:

- The system asks the user to enter the course code of the course that needs to be deleted.
- The program uses the findCourseIndex() method (linear search) to locate the course in the ArrayList.
- If the course exists, the program retrieves its index for further action.

8.4.2 DISPLAY COURSE ATTRIBUTES WITH CONFIRMATION MESSAGE

```
165                 Course c = courses.get(index);
166                 System.out.println("Course found:");
167                 c.displayCourse();
168
169                 System.out.print("Confirm deletion? (Y/N): ");
170                 String confirm = sc.nextLine();
```

```
Course found:
Course Name     : SOFTWARE ENGINEERING
Course Code     : CSEB3123
Credit Hour     : 3
Summary         : SDLC
MS Teams Link   : https://teams.microso
-------------------------------------
Confirm deletion? (Y/N): Y
```

Explanation:

- If the course is found, the system displays the complete course attributes using the displayCourse() method.
- This allows the user to verify the correct course before deleting it.
- The program then asks for confirmation (Y/N) before proceeding with deletion.

8.4.3 DISPLAY MESSAGE FOR UNSUCCESSFUL SEARCH

```
180         } else {
181             System.out.println("Course not found.");
182         }
183     }
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 4
Enter Course Code to delete: CSEB5113
Course not found.
```

Explanation:

- If the entered course code does not exist, the system displays the message "Course not found."
- This prevents the program from attempting to delete a non-existing course.

8.5 VIEW ALL COURSE FUNCTION DEVELOPMENT

8.5.1 IMPLEMENT A 'VIEW ALL' FUNCTION TO DISPLAY ALL COURSES

```
46                  case 5: viewAllCourses(); break;
```

```
--- COURSE PROFILE SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
0. Exit
Choose an option: 5
```

Explanation:

- The system provides a menu option (5 – View All Courses) that acts like a "View All" button.
- When the user selects this option, the program calls the viewAllCourses() method.
- This method displays all courses currently stored in the system.

8.5.2 METHOD TO DISPLAY COURSE ATTRIBUTES IN ORGANIZED FORMAT

```
188     private void viewAllCourses() {
189         if (courses.isEmpty()) {
190             System.out.println("No courses available.");
191         } else {
192             System.out.println("\n--- All Courses ---");
193             for (Course c : courses) {
194                 c.displayCourse();
195             }
196         }
197     }
```

```
--- All Courses ---
Course Name     : SOFTWARE SECURITY
Course Code     : CSEB5113
Credit Hour     : 3
Summary         : SECURITY
MS Teams Link   : https://teams.microsoft.com/l/tea
---------------------------------------
Course Name     : SOFTWARE CONSTRUCTION & METHOD
Course Code     : CSEB5223
Credit Hour     : 3
Summary         : GITHUB AND CODING
MS Teams Link   : https://teams.microsoft.com/l/tea
---------------------------------------
```

Explanation:

- This method displays all courses stored in the ArrayList.
- It first checks whether the list is empty.
- If courses exist, the system loops through the list and calls displayCourse() for each course.
- The displayCourse() method ensures the information is shown in a clear and organized format.

9.0 STUDENT CLASS

9.1 STUDENT CLASS DESIGN

- **Attributes**:

```
 * This class includes:
 * - Constructor to create a Student object
 * - Getter methods to access attributes
 * - Setter methods to modify editable attributes
 * - displayStudent method to show student information in a clear format
 */
public class Student {

    // Attributes
    private String studentID;   // Unique student identifier, cannot be changed after creation
    private String firstName;   // Student's first name
    private String lastName;    // Student's last name
    private String email;       // Student's email address
    private String phone;       // Student's phone number
```

  - studentID (String) – Unique, immutable identifier for each student.
  - firstName (String) – Student's first name.
  - lastName (String) – Student's last name.
  - email (String) – Student's email address.
  - phone (String) – Student's contact number.

- **Methods**:
  - **Getters**: Allow other classes (e.g., CourseManager) to access private attributes.
  - **Setters**: Allow modification of firstName, lastName, email, phone but not studentID.
  - **displayStudent()**: Prints all student attributes in a clear and organized format.

9.2 ADD STUDENT FUNCTION

The system allows users to manually enter student information.

```
    // Validate all inputs at once
    if (id.isBlank() || fname.isBlank() || lname.isBlank() || email.isBlank() || phone.isBlank()) {
        System.out.println("Student not added! All fields must be filled.");
    } else {
        // Only add if all fields are valid
        students.add(new Student(id, fname, lname, email, phone));
        System.out.println("Student added successfully!");
    }
}
```

- User enters Student ID, First Name, Last Name, Email, and Phone Number.
- The system validates whether any field is empty.
- If any field is blank:
- "Student not added! All fields must be filled."
- If all fields are filled:
- "Student added successfully!"
- The student object is then stored in an **ArrayList**, allowing multiple student records to be stored dynamically.

- **Successfully added:**

```
--- SLMS SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
6. Add Student
7. Search Student
8. Edit Student
9. Delete Student
10. View All Students
0. Exit
Choose an option: 6
Enter Student ID: BSW01085928
Enter First Name: Wan Nur
Enter Last Name: Sabrina
Enter Email: sabrina@gmail.com
Enter Phone: 0123456678
Student added successfully!
```

- **Students fail to fill all fields:**

```
Choose an option: 6
Enter Student ID:
Enter First Name: wan
Enter Last Name: sabrina
Enter Email: sabrina@gmail.com
Enter Phone: 0165766482
Student not added! All fields must be filled.
```

9.3 SEARCH STUDENT FUNCTION

The search function allows users to find a student using **Student ID**.

```
private void searchStudent() {
    System.out.print("Enter Student ID to search: ");
    String id = sc.nextLine();

    int index = findStudentIndex(id); // search for student

    if (index != -1) {
        System.out.println("Student found:");
        students.get(index).displayStudent(); // Show student details
    } else {
        System.out.println("Student not found."); // not found
    }
}
```

- User enters a student ID.
- The system uses linear search with equalsIgnoreCase() → search is case-insensitive.
- If found → displays all student information using displayStudent().
- If not found → displays: "Student not found."
- Search is case-insensitive, so entering lowercase/uppercase IDs works.
- Example: searching for bsw01085998 successfully finds the student.

- **Student was found:**

```
Choose an option: 6
Enter Student ID: BSW01085998
Enter First Name: Ilyas
Enter Last Name: Ahmad
Enter Email: ilyass@gmail.com
Enter Phone: 0198765543
Student added successfully!
 7. Search Student
 8. Edit Student
 9. Delete Student
 10. View All Students
 0. Exit
 Choose an option: 7
 Enter Student ID to search: bsw01085998
 Student found:
 Student ID   : BSW01085998
 Name         : Ilyas Ahmad
 Email        : ilyass@gmail.com
 Phone        : 0198765543
 --------------------------------------
```

- **Student was not found:**

```
Choose an option: 7
Enter Student ID to search: BSW01098765
Student not found.
```

9.4 EDIT STUDENT FUNCTION

- User enters Student ID.
- The system performs a **linear search** to locate the student.

If found:

The user can edit:

- FirstName
- LastName
- Email
- phone

The **studentID cannot be edited** because it is a unique identifier.

After editing, the system displays the **updated student information** to confirm the changes.

If student is not found:

```
Choose an option: 7
Enter Student ID to search: SW03
Student not found.
```

Outputs:

- **Before Edit:**

```
Choose an option: 6
Enter Student ID: BSW01085973
Enter First Name: Hidayah
Enter Last Name: Sarwani
Enter Email: dayah@gmail.com
Enter Phone: 0167542987
Student added successfully!
```

- **After Edited:**

```
Choose an option: 8
Enter Student ID to edit: BSW01085973
New First Name: Nur Dayah
New Last Name: Sarwani
New Email: dayah@gmail.com
New Phone: 0167542987
Student updated successfully.
Student ID   : BSW01085973
Name         : Nur Dayah Sarwani
Email        : dayah@gmail.com
Phone        : 0167542987
--------------------------------------
```

- **Student Not Found:**

```
Choose an option: 8
Enter Student ID to edit: BSW01085974
Student not found.
```

Good Practice Observed:

- Clear separation of ID (immutable) and editable fields.
- Updated info displayed for verification.

9.5 DELETE STUDENT FUNCTION

```
private void deleteStudent() {
    System.out.print("Enter Student ID to delete: ");
    String id = sc.nextLine();

    int index = findStudentIndex(id);

    if (index != -1) {
        Student s = students.get(index);
        System.out.println("Student found:");
        s.displayStudent(); // show info before deletion

        System.out.print("Confirm deletion? (Y/N): ");
        String confirm = sc.nextLine();

        if (confirm.equalsIgnoreCase("Y")) {
            students.remove(index); // delete student
            System.out.println("Student deleted successfully!");
            viewAllStudents(); // Show updated list
        } else {
            System.out.println("Deletion cancelled.");
        }
    } else {
        System.out.println("Student not found.");
    }
}
```

- User enters student ID.
- If ID not found → "Student not found."
- If found → system displays student info and asks for confirmation (Y/N).
- User confirms deletion → student removed from list.

Outputs:

- **Student Found and Deleted:**

```
--- SLMS SYSTEM ---
1. Add Course
2. Search Course
3. Edit Course
4. Delete Course
5. View All Courses
6. Add Student
7. Search Student
8. Edit Student
9. Delete Student
10. View All Students
0. Exit
Choose an option: 9
Enter Student ID to delete: BSW01085973
Student found:
Student ID    : BSW01085973
Name          : Nur Dayah Sarwani
Email         : dayah@gmail.com
Phone         : 0167542987
--------------------------------------
Confirm deletion? (Y/N): Y
Student deleted successfully!

--- All Students ---
Student ID    : BSW01085928
Name          : Wan Nur Sabrina
Email         : sabrina@gmail.com
Phone         : 0123456678
--------------------------------------
```

- **Student ID Not Found:**

```
Enter Student ID to delete: bsw01098786
Student not found.
```

Good Practices Observed:

- Confirms deletion to prevent accidental removal.
- Case-insensitive confirmation input (y/Y) works.
- Clear feedback for both success and failure.

9.6 VIEW ALL STUDENTS FUNCTION

The **View All Students** function displays all student records stored in the system.

Implementation:

- The system loops through the **ArrayList** of students.
- The **displayStudent()** method is called for each student.

Before displaying, the system checks whether the list is empty to prevent errors.

This ensures student information is shown in a **clear and organized format**.

```
public void displayStudent(){
    System.out.println("Student ID   : " + studentID);
    System.out.println("Name         : " + firstName + " " + lastName);
    System.out.println("Email        : " + email);
    System.out.println("Phone        : " + phone);
    System.out.println("-------------------------------------");
}
}
```

OUTPUT:

```
--- All Students ---
Student ID   : BSW01085928
Name         : Wan Nur Sabrina
Email        : sabrina@gmail.com
Phone        : 0123456678
-------------------------------------
Student ID   : BSW01085973
Name         : Hidayah Sarwani
Email        : dayah@gmail.com
Phone        : 0167542987
-------------------------------------

Student ID   : BSW01085998
Name         : Ilyas Ahmad
Email        : ilyass@gmail.com
Phone        : 0198765543
-------------------------------------
```

Good Practices Observed:

- **Reusability**: Uses displayStudent() instead of repeating print statements.

- **Validation**: Handles empty list gracefully.
- Checks if the list is empty.

9.7 EXTRA VALIDATION

- Lowercase IDs still work due to equalsIgnoreCase().
- Example: Searching for bsw01085976 still finds the student.

LOWERCASE ID:

```
Enter Student ID to search: bsw01085998
Student found:
Student ID    : BSW01085998
Name          : Ilyas Ahmad
Email         : ilyass@gmail.com
Phone         : 0198765543
--------------------------------------
```

LOWERCASE CONFIRMATION:

```
Choose an option: 9
Enter Student ID to delete: bsw01085998
Student found:
Student ID    : BSW01085998
Name          : Ilyas Irfan
Email         : ilyass@gmail.com
Phone         : 0198765563
--------------------------------------
Confirm deletion? (Y/N): y
Student deleted successfully!

--- All Students ---
Student ID    : BSW01085928
Name          : Wan Nur Sabrina
Email         : sabrina@gmail.com
Phone         : 0123456678
--------------------------------------
```

Good Practices:

- All fields validated after input.

- User-friendly: all fields input first without interruption.
- Clear feedback: success/failure at the end.
- Encapsulation: Student attributes remain private, added via constructor.

10.0 COURSEMANAGER CLASS

10.1 COURSEMANAGER CLASS DESIGN

- **Class declaration**:

```
34    class CourseManager {
```

Explanation:

- This is the main controller class of the system.
- It manages:
  - Course data
  - Student data
  - Relationship between students and courses
- It acts as the core logic handler for the SLMS (Student Learning Management System).

- **Attributes**:

```
35      // Stores last searched student ID (for history feature)
36      private String lastSearchedStudent = "";
37
38      // Stores last searched course code (for history feature)
39      private String lastSearchedCourse = "";
40
41      // ArrayList to store all courses. ArrayList allows dynamic resizing.
42      private ArrayList<Course> courses = new ArrayList<>();
43
44      // ArrayList to store all students. Dynamic insertion and deletion possible.
45      private ArrayList<Student> students = new ArrayList<>();
46
47      // Stores search history for API caching feature
48      private ArrayList<String> searchCache = new ArrayList<>();
49
50      /// Relationship matrix:
51      // enrollment[studentIndex][courseIndex] = true if student is enrolled
52      private boolean[][] enrollment = new boolean[100][100];
```

Explanation:

1. lastSearchedStudent
   - Stores the last searched student ID
   - Used for search history feature and suggestion system

2. lastSearchedCourse
   - Stores the last searched course code
   - Helps improve:
   - User experience
   - Quick recall of previous searches

3. courses
    - Stores all Course objects
    - Uses ArrayList because dynamic size (can grow/shrink), easy to add, delete and edit

4. students
    - Stores all Student objects
    - Supports dynamic insertion and efficient management of student records

5. searchCache
    - Stores search history
    - Acts as a simple caching mechanism
    - Improves performance and reusability of search data

6. enrollment
    - A 2D array (matrix) to manage relationships
    - Represents:

```
enrollment[sIndex][cIndex]
```

    - Value meaning:
        - `true` → student is enrolled in course
        - `false` → not enrolled
    - This is called a relationship matrix
    - Efficient for fast lookup and relationship tracking

10.2 RELATIONSHIP MANAGEMENT FUNCTION DEVELOPMENT

10.2.1 ASSIGN COURSE TO STUDENT FUNCTION

This method assigns a course to a student by creating a relationship between them.

```
613     private void assignCourseToStudent() {
614         System.out.print("Enter Student ID: ");
615         String studentId = sc.nextLine();
616         int sIndex = findStudentIndex(studentId);
617 
618         if (sIndex == -1) {
619             System.out.println("Error: Student not found.");
620             return;
621         }
622 
623         System.out.print("Enter Course Code: ");
624         String courseCode = sc.nextLine();
625         int cIndex = findCourseIndex(courseCode);
626 
627         if (cIndex == -1) {
628             System.out.println("Error: Course not found.");
629             return;
630         }
631 
632         if (enrollment[sIndex][cIndex]) {
633             System.out.println("Error: Student already enrolled in this course.");
634             return;
635         }
636 
637         // if NOT enrolled yet → assign
638         enrollment[sIndex][cIndex] = true;
639         System.out.println("Success: Course assigned to student.");
640     }
```

The user enters a student ID and a course code as input.

```
Choose an option: 11
Enter Student ID: sw01083600
Enter Course Code: CISB3323
Success: Course assigned to student.
```

The program first checks whether the student exists using the findStudentIndex() method.

```
589     private int findStudentIndex(String id) {
590         for (int i = 0; i < students.size(); i++) {
591             // equalsIgnoreCase: allows case-insensitive comparison
592             if (students.get(i).getStudentID().equalsIgnoreCase(id)) {
593                 return i; // student found, display student's information
594             }
595         }
```

If the student is not found, an error message is displayed and the process stops.

```
596         // -1 indicates student not found
597         return -1;
598     }
```

Next, the program checks whether the course exists using the findCourseIndex() method.

```
373     private int findCourseIndex(String code) {
374         for (int i = 0; i < courses.size(); i++) {
375             // equalsIgnoreCase: ignores uppercase/lowercase differences
376             if (courses.get(i).getCourseCode().equalsIgnoreCase(code)) {
377                 return i; // Course found
378             }
379         }
```

If the course is not found, an error message is displayed.

```
380         // -1 indicates course not found
381         return -1;
382     }
```

The system then verifies whether the student is already enrolled in the selected course by checking the enrollment matrix.

```
51      // enrollment[studentIndex][courseIndex] = true if student is enrolled
52      private boolean[][] enrollment = new boolean[100][100];
```

If the student is already enrolled, the system prevents duplication and shows an error message.

```
Choose an option: 11
Enter Student ID: SW01083600
Enter Course Code: CISB3323
Error: Student already enrolled in this course.
```

If all validations pass, the system updates the enrollment matrix by setting the value to true.

This indicates that the student is successfully assigned to the course.

10.2.2 LIST COURSES BY STUDENT FUNCTION

This method displays all courses assigned to a specific student.

```
649     private void listCoursesByStudent() {
650         System.out.print("Enter Student ID: ");
651         String id = sc.nextLine();
652         int sIndex = findStudentIndex(id);
653 
654         if (sIndex == -1) {
655             System.out.println("Error: Student not found.");
656             return;
657         }
658 
659         boolean found = false;
660 
661         for (int j = 0; j < courses.size(); j++) {
662             if (enrollment[sIndex][j]) {
663                 courses.get(j).displayCourse();
664                 found = true;
665             }
666         }
667 
668         if (!found) {
669             System.out.println("This student has no courses.");
670         }
671     }
```

The user enters a student ID as input.

```
Choose an option: 12
Enter Student ID: sw01083600
```

The program uses the findStudentIndex() method to locate the student.

```
589     private int findStudentIndex(String id) {
590         for (int i = 0; i < students.size(); i++) {
591             // equalsIgnoreCase: allows case-insensitive comparison
592             if (students.get(i).getStudentID().equalsIgnoreCase(id)) {
593                 return i; // student found, display student's information
594             }
595         }
596         // -1 indicates student not found
597         return -1;
598     }
```

If the student does not exist, an error message is displayed.

```
Choose an option: 12
Enter Student ID: sw01083400
Error: Student not found.
```

If the student exists, the program loops through all courses using a for loop.

For each course, the system checks the enrollment matrix to determine whether the student is enrolled.

If the value is true, the course details are displayed using the displayCourse() method.

```
Choose an option: 12
Enter Student ID: sw01083600
Course Name  : HCI
Course Code  : CISB3323
Credit Hour  : 3
Summary      : Human Computer Interaction
MS Teams Link: teams.hci
-------------------------------------
Course Name  : AI
Course Code  : CISB3433
Credit Hour  : 3
Summary      : Artificial Intelligence
MS Teams Link: teams.ai
-------------------------------------
Course Name  : SCM
Course Code  : CISB3533
Credit Hour  : 3
Summary      : Software Construction Management
MS Teams Link: teams.scm
-------------------------------------
```

A boolean variable is used to track whether any courses are found.

If no courses are assigned, the system displays the message “This student has no courses.”

```
Choose an option: 12
Enter Student ID: sw01083500
This student has no courses.
```

10.2.3 LIST STUDENTS BY COURSE FUNCTION

This method displays all students enrolled in a specific course.

```
681     private void listStudentsByCourse() {
682         System.out.print("Enter Course Code: ");
683         String code = sc.nextLine();
684         int cIndex = findCourseIndex(code);
685 
686         if (cIndex == -1) {
687             System.out.println("Error: Course not found.");
688             return;
689         }
690 
691         boolean found = false;
692 
693         for (int i = 0; i < students.size(); i++) {
694             if (enrollment[i][cIndex]) {
695                 students.get(i).displayStudent();
696                 found = true;
697             }
698         }
699 
700         if (!found) {
701             System.out.println("No students enrolled in this course.");
702         }
703     }
```

The user enters a course code as input.

```
Choose an option: 13
Enter Course Code: cisb3433
```

The program calls the findCourseIndex() method to locate the course.

```
373     private int findCourseIndex(String code) {
374         for (int i = 0; i < courses.size(); i++) {
375             // equalsIgnoreCase: ignores uppercase/lowercase differences
376             if (courses.get(i).getCourseCode().equalsIgnoreCase(code)) {
377                 return i; // Course found
378             }
379         }
```

If the course is not found, an error message is displayed.

```
Choose an option: 13
Enter Course Code: csnb5233
Error: Course not found.
```

If the course exists, the program loops through all students using a for loop.

For each student, the system checks the enrollment matrix to verify if the student is enrolled in the course.

If the value is true, the student details are displayed using the displayStudent() method.

```
Choose an option: 13
Enter Course Code: cisb3433
Student ID   : SW01083600
Name         : Abu Ali
Email        : abu@gmail.com
Phone        : 019-9099876
--------------------------------------
Student ID   : sw01083500
Name         : aminah ali
Email        : aminah@gmail.com
Phone        : 01988776655
--------------------------------------
Student ID   : sw01083700
Name         : said ali
Email        : said@gmail.com
Phone        : 01344556677
--------------------------------------
```

A boolean variable is used to check whether any students are enrolled.

If no students are found, the system displays the message “No students enrolled in this course.”

```
Choose an option: 13
Enter Course Code: csnb3333
No students enrolled in this course.
```

10.2.4 SUGGEST LAST SEARCH FUNCTION

This method displays the most recent search history of the user.

```
713     private void suggestLastSearch() {
714         if (lastSearchedStudent.isEmpty() && lastSearchedCourse.isEmpty()) {
715             System.out.println("No search history yet.");
716         } else {
717             System.out.println("Last searched student: " + lastSearchedStudent);
718             System.out.println("Last searched course: " + lastSearchedCourse);
719         }
720     }
```

It shows the last searched student ID and course code.

The system first checks whether both values are empty.

If no search has been performed, the system displays “No search history yet.”

```
Choose an option: 14
No search history yet.
```

Otherwise, it displays the last searched student and course.

```
Choose an option: 14
Last searched student: sw01083500
Last searched course: csnb3333
```

This feature improves user experience by allowing quick reference to previous searches.

10.2.5 AUTO SUGGEST FUNCTION

This method provides an auto-suggestion feature based on user input.

```
735    private void autoSuggest(String input) {
736        System.out.println("Suggestions:");
737
738        boolean found = false;
739
740        for (Course c : courses) {
741            if (c.getCourseCode().toLowerCase().contains(input.toLowerCase()) ||
742                c.getCourseName().toLowerCase().contains(input.toLowerCase())) {
743
744                System.out.println("- Course: " + c.getCourseCode());
745                found = true;
746            }
747        }
748
749        if (!found) {
750            System.out.println("No suggestions found.");
751        }
752    }
```

The user enters a keyword to search.

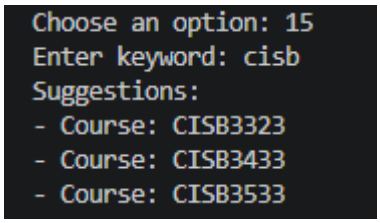
```
Choose an option: 15
Enter keyword: csnb
Suggestions:
- Course: csnb3333
```

The system loops through all courses in the courses ArrayList.

For each course, the program compares the keyword with the course code and course name.

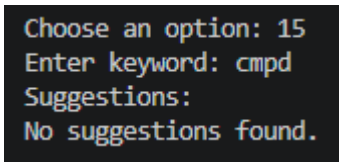
The comparison is done using toLowerCase() to ensure case-insensitive matching.

If a match is found, the course code is displayed as a suggestion.



A boolean variable is used to track whether any matching results exist.

If no matches are found, the system displays “No suggestions found.”



This feature enhances usability by helping users quickly locate relevant courses.

11.0 ANALYSIS OF SOFTWARE CONSTRUCTION BEST PRACTICES AND CHALLENGES

11.1 BEST PRACTICES APPLIED

Several software construction best practices have been successfully applied in the development of the Course–Student Relationship and Middleware API module.

Firstly, modularity and separation of concerns are clearly implemented. The system is divided into three main classes, namely Course, Student, and CourseManager. Each class is responsible for a specific functionality, which improves code organization, readability, and maintainability.

Secondly, encapsulation is properly applied. All attributes in the Course and Student classes are declared as private, and access is controlled through getter and setter methods. This ensures data integrity and prevents unauthorized modification of important attributes such as student ID and course code.

In addition, input validation and error handling are consistently implemented throughout the system. The program validates user inputs, prevents duplicate entries, and handles invalid operations such as searching for non-existent records or assigning duplicate enrollments. This improves system reliability and reduces runtime errors.

Another important practice is the use of appropriate data structures. The combination of ArrayList and a 2D array (boolean matrix) enables efficient storage and management of student, course, and relationship data. This approach supports dynamic data handling while maintaining a clear relationship mapping.

Furthermore, code documentation and commenting are well applied. Clear and informative comments are included in the code to explain the purpose of classes, methods, and logic flow. This enhances code readability and makes it easier for future developers to understand and maintain the system.

Lastly, the implementation of basic middleware API features, such as caching and auto-suggestion, demonstrates an understanding of enhancing user experience through smarter data handling. These features reduce repetitive input and improve interaction efficiency.

11.2 CHALLENGES FACED

Despite the successful implementation, several challenges were encountered during the development of this module.

One of the main challenges was managing the course–student relationship using a 2D array. Since the array uses fixed size and index-based mapping, additional care was required when adding or deleting students and courses to ensure that the relationship data remains consistent and does not cause misalignment.

Another challenge was ensuring data consistency during deletion operations. When a student or course is removed, the system must also update the enrollment matrix accordingly. Implementing this correctly required careful handling of index shifting and clearing of relationships.

The implementation of error handling and edge cases was also challenging. The system needed to consider multiple scenarios, such as duplicate enrollments, missing data, and empty results. Ensuring that all these cases are properly handled without breaking the program logic required thorough testing and validation.

In addition, developing the auto-suggestion and caching features required extra logic beyond basic CRUD operations. Designing a simple yet functional approach to simulate middleware API behavior using local data structures was a new and challenging task.

Finally, maintaining code organization and readability while handling multiple features in a single system was also a challenge. Proper structuring of methods and consistent coding style were necessary to avoid confusion and ensure the system remains maintainable.

11.3 CONCLUSION

In conclusion, the module demonstrates the application of key software construction best practices, including modular design, encapsulation, input validation, and proper data structure usage. Although several challenges were encountered, they were successfully addressed, resulting in a functional, reliable, and well-structured system.

12.0 EVALUATION OF STUDENT PROFILE MODULE

The Course Student Relationship and Middleware API module successfully fulfills all the requirements specified in the Lab 7–8 evaluation criteria. The system effectively manages relationships between students and courses using a 2D array (boolean matrix), where each element represents the enrollment status between a student and a course. This approach ensures efficient data handling, avoids redundancy, and provides a clear mapping between entities.

All required relationship management functions are correctly implemented. The assignCourseToStudent() method allows users to enroll students into courses while preventing duplicate assignments. The listCoursesByStudent() function displays all courses enrolled by a particular student, whereas the listStudentsByCourse() function retrieves all students registered under a specific course. These operations demonstrate proper handling of many-to-many relationships and ensure data consistency within the system.

The program also demonstrates strong robustness through comprehensive error handling and edge case management. The system validates the existence of students and courses before performing operations, handles cases where students have no assigned courses or courses have no enrolled students, and prevents duplicate enrollments using the enrollment matrix. Additionally, meaningful and user-friendly error messages are provided, which improves system usability and reduces the likelihood of user mistakes.

The middleware API features are implemented effectively to enhance system functionality. A caching mechanism is introduced using an ArrayList to store search history, allowing the system to remember previously accessed data. The suggestLastSearch() function enables quick retrieval of recent searches, while the autoSuggest() function provides real-time input suggestions based on partial keywords entered by the user. These features simulate basic API behavior and improve overall user experience by reducing repetitive input.

In terms of software construction principles, the system demonstrates good modularity and separation of concerns. The Course, Student, and CourseManager classes are well-organized, with each class handling its specific responsibility. Encapsulation is properly applied by restricting direct access to attributes and using getter and setter methods. The use of structured programming, meaningful method decomposition, and clear control flow enhances code readability and maintainability.

Furthermore, input validation is consistently applied throughout the system to ensure data integrity. The program checks for invalid inputs such as empty fields, incorrect data types, and duplicate identifiers. This contributes to a more reliable and stable system.

One of the key strengths of this module is its successful integration with previous modules. The combination of student management, course management, and relationship handling within a single menu-driven interface demonstrates a well-integrated system architecture. This integration ensures smooth interaction between components and reflects a practical implementation of a real-world student management system.

Overall, the module is well-designed, functionally complete, and meets all evaluation criteria, including data structure implementation, relationship management, modularity, error handling, middleware API integration, and user interface functionality. The system is user-friendly, scalable, and demonstrates a strong understanding of software construction best practices.

13.0 APPENDIX

13.1 Course.java

```java
/**
 * File: Course.java
 * Author: Team SLMS (Group 3)
 * Description: Handles individual course data and provides methods to display it.
 * Date: 02 Mar 2026
 */

package com.mycompany.courseprofilesystem;

/**
 * Class: Course
 * Description: Represents a course profile in the SLMS system.
 * Stores course information such as name, code, credit hour, summary,
 * and MS Teams link.
 */

public class Course {
    // Attributes available in the course
    private String courseName;
    private String courseCode; // cannot be edited after creation
    private int creditHour;
    private String summary;
    private String msTeamsLink;

    // Constructor for the course
    public Course(String courseName, String courseCode, int creditHour, String summary, String msTeamsLink) {
        this.courseName = courseName;
        this.courseCode = courseCode;
        this.creditHour = creditHour;
        this.summary = summary;
        this.msTeamsLink = msTeamsLink;
    }

    // Getters and Setters for the required course's attributes
    public String getCourseName() { return courseName; }
    public void setCourseName(String courseName) { this.courseName = courseName; }

    public String getCourseCode() { return courseCode; }

    public int getCreditHour() { return creditHour; }
    public void setCreditHour(int creditHour) { this.creditHour = creditHour; }

    public String getSummary() { return summary; }
    public void setSummary(String summary) { this.summary = summary; }

    public String getMsTeamsLink() { return msTeamsLink; }
    public void setMsTeamsLink(String msTeamsLink) { this.msTeamsLink = msTeamsLink; }

    // Display all course's attributes
    public void displayCourse() {
        System.out.println("Course Name  : " + courseName);
        System.out.println("Course Code  : " + courseCode);
        System.out.println("Credit Hour  : " + creditHour);
        System.out.println("Summary      : " + summary);
        System.out.println("MS Teams Link: " + msTeamsLink);
        System.out.println("-------------------------------------");
    }
}
```

13.2 Student.java

```java
/**
 * File: Student.java
 * Author: Team SLMS (Group 3)
 * Description: Handles individual student data and provides methods to display it.
 * Date: 17 Mar 2026
 */

package com.mycompany.courseprofilesystem;

/**
 * Class: Student
 * Description:
 * This class represents a Student object in the SLMS system.
 *
 * Each student has all these attributes:
 * - studentID: unique identifier (cannot be edited)
 * - firstName
 * - lastName
 * - email
 * - phone
 *
 * This class includes:
 * - Constructor to create a Student object
 * - Getter methods to access attributes
 * - Setter methods to modify editable attributes
 * - displayStudent method to show student information in a clear format
 */
public class Student {

    // Student's attributes
    private String studentID;   // Unique student identifier, cannot be changed after creation
    private String firstName;   // Student's first name
    private String lastName;    // Student's last name
    private String email;       // Student's email address
    private String phone;       // Student's phone number


    /**
     * Constructor: Student
     * Purpose:
     * - Initialize a new Student object with all required attributes
     * - studentID is passed in constructor because it is unique and must not be changed later
     */
    public Student(String studentID, String firstName, String lastName, String email, String phone){
        this.studentID = studentID;
        this.firstName = firstName;
        this.lastName = lastName;
        this.email = email;
        this.phone = phone;
    }

    /**
     * Getter methods:
     * These allow other classes (e.g., CourseManager) to access student attributes.
     * We do not allow direct access to private attributes for data encapsulation.
     *
     * Encapsulation ensures data safety and control over how attributes are accessed or modified.
     */
```

```
59
60          public String getStudentID(){
61              return studentID; // StudentID is unique and immutable, so only getter is provided
62          }
63
64          public String getFirstName(){
65              return firstName;
66          }
67
68          public String getLastName(){
69              return lastName;
70          }
71
72          public String getEmail(){
73              return email;
74          }
75
76          public String getPhone(){
77              return phone;
78          }
79
80          /**
81           * Setter methods:
82           * These allow modification of attributes except studentID.
83           * Why no setter for studentID?
84           * - studentID is the unique identifier.
85           * - Changing it could cause duplicate conflicts or break search/delete operations.
86           */
87
88          public void setFirstName(String firstName){
89              this.firstName = firstName;
90          }
91
92          public void setLastName(String lastName){
93              this.lastName = lastName;
94          }
95
96          public void setEmail(String email){
97              this.email = email;
98          }
99
100         public void setPhone(String phone){
101             this.phone = phone;
102         }
103
104         /**
105          * Method: displayStudent
106          * Purpose:
107          * - To show all student attributes in a clear and organized format
108          * - Used after search, edit, or delete operations in CourseManager
109          *
110          * Why use a separate display method?
111          * - To avoid repeating System.out.println code in multiple places
112          * - Improves code reusability and readability
113          */
114
115         public void displayStudent(){
116             System.out.println("Student ID   : " + studentID);
117             System.out.println("Name         : " + firstName + " " + lastName);
118             System.out.println("Email        : " + email);
119             System.out.println("Phone        : " + phone);
120             System.out.println("-----------------------------------");
121         }
122     }
```

13.3 CourseManager.java

```java
/**
 * File: CourseManager.java
 * Author: Team SLMS (Group 3)
 * Description: Main program to manage course profiles.
 * Features: Add, Search, Edit, Delete, View All courses.
 */

package com.mycompany.courseprofilesystem;

import java.util.ArrayList;
import java.util.Scanner;

/**
 * Class: CourseManager
 *
 * This class manages both Course Profile and Student Profile modules
 * in the SLMS (Student Learning Management System).
 *
 * Users can:
 * - Add, search, edit, delete, and view courses
 * - Add, search, edit, delete, and view students
 *
 * ArrayLists are used for dynamic storage of Course and Student objects.
 * Scanner is used for user input.
 */
class CourseManager {

    // ArrayList to store all courses. ArrayList allows dynamic resizing.
    private ArrayList<Course> courses = new ArrayList<>();

    // ArrayList to store all students. Dynamic insertion and deletion possible.
    private ArrayList<Student> students = new ArrayList<>();

    // Scanner to get user input
    private Scanner sc = new Scanner(System.in);

    /**
     * Main method: program entry point
     */
    public static void main(String[] args) {
        CourseManager manager = new CourseManager();
        manager.run();  // Start the interactive menu
    }

    /**
     * Method: run()
     *
     * This is the main menu loop. It uses a do-while loop.
     *
     * Why use do-while?
     * - Ensures menu displays at least once before checking exit.
     * - Loop continues until user chooses 0 (Exit).
     */
```

```
54      public void run() {
55          int choice;
56          do {
57              // Display menu options
58              System.out.println("\n--- SLMS SYSTEM ---");
59              System.out.println("1. Add Course");
60              System.out.println("2. Search Course");
61              System.out.println("3. Edit Course");
62              System.out.println("4. Delete Course");
63              System.out.println("5. View All Courses");
64              System.out.println("6. Add Student");
65              System.out.println("7. Search Student");
66              System.out.println("8. Edit Student");
67              System.out.println("9. Delete Student");
68              System.out.println("10. View All Students");
69              System.out.println("0. Exit");
70              System.out.print("Choose an option: ");
71
72              /**
73               * Input validation using while loop.
74               * - Checks if the user entered an integer.
75               * - If input is invalid (like letters), the user need to prompt again.
76               * - sc.next() clears the invalid input.
77               */
78              while (!sc.hasNextInt()) {
79                  System.out.print("Invalid input! Enter a number: ");
80                  sc.next();
81              }
82              choice = sc.nextInt();
83              sc.nextLine();  // consume newline
84
85              // Switch case to execute chosen operation
86              switch (choice) {
87                  case 1: addCourse(); break;
88                  case 2: searchCourse(); break;
89                  case 3: editCourse(); break;
90                  case 4: deleteCourse(); break;
91                  case 5: viewAllCourses(); break;
92                  case 6: addStudent(); break;
93                  case 7: searchStudent(); break;
94                  case 8: editStudent(); break;
95                  case 9: deleteStudent(); break;
96                  case 10: viewAllStudents(); break;
97                  case 0: System.out.println("Exiting program."); break;
98                  default: System.out.println("Invalid choice!");
99              }
100
101         } while (choice != 0); // Loop until user enters 0
102     }
```

```
104         /**
105          * Method: addCourse()
106          * Adds a new course after valid inputs.
107          */
108         private void addCourse() {
109             System.out.print("Enter Course Name: ");
110             String name = sc.nextLine();
111 
112             String code;
113 
114             // Ensure unique course code
115             while (true) {
116                 System.out.print("Enter Course Code: ");
117                 code = sc.nextLine();
118 
119                 // findCourseIndex() searches if code already exists
120                 if (findCourseIndex(code) != -1) {
121                     System.out.println("Course code already exists! Enter another unique code.");
122                 } else break; // Exit loop when course code is unique
123             }
124 
125             int credit;
126             // Validate credit hour input: integer and > 0
127             while (true) {
128                 System.out.print("Enter Credit Hour: ");
129                 if (sc.hasNextInt()) {
130                     credit = sc.nextInt();
131                     sc.nextLine();
132                     if (credit > 0) break;
133                     else System.out.println("Credit hour must be positive!");
134                 } else {
135                     System.out.println("Invalid input! Enter a number.");
136                     sc.next();
137                 }
138             }
139 
140             System.out.print("Enter Course Summary: ");
141             String summary = sc.nextLine();
142 
143             System.out.print("Enter MS Teams Link: ");
144             String link = sc.nextLine();
145 
146             // Validate all inputs at once
147             if (name.isBlank() || code.isBlank() || credit.isBlank() || summary.isBlank() || link.isBlank()) {
148                 System.out.println("Course not added! All fields must be filled.");
149             } else {
150                 // Only add if all fields are valid
151                 courses.add(new Course(name, code, credit, summary, link));
152                 System.out.println("Course added successfully!");
153             }
```

```
151         /**
152          * Method: searchCourse()
153          * Search for a course by course code.
154          */
155         private void searchCourse() {
156             System.out.print("Enter Course Code to search: ");
157             String code = sc.nextLine();
158 
159             int index = findCourseIndex(code);
160 
161             if (index != -1) {
162                 System.out.println("Course found:");
163                 courses.get(index).displayCourse(); // Display course info
164             } else {
165                 System.out.println("Course not found.");
166             }
167         }
168 
```

```
169         /**
170          * Method: editCourse()
171          * Edit existing course details (except code).
172          */
173         private void editCourse() {
174             System.out.print("Enter Course Code to edit: ");
175             String code = sc.nextLine();
176             int index = findCourseIndex(code);
177 
178             if (index != -1) {
179                 Course c = courses.get(index);
180                 System.out.println("Editing Course: " + c.getCourseCode());
181 
182                 System.out.print("New Course Name: ");
183                 c.setCourseName(sc.nextLine());
184 
185                 int credit;
186                 // Validate input to new credit hour: integer and > 0
187                 while (true) {
188                     System.out.print("New Credit Hour: ");
189                     if (sc.hasNextInt()) {
190                         credit = sc.nextInt();
191                         sc.nextLine();
192                         if (credit > 0) break;
193                         else System.out.println("Credit hour must be positive!");
194                     } else {
195                         System.out.println("Invalid input! Enter a number.");
196                         sc.next();
197                     }
198                 }
199                 c.setCreditHour(credit);
200 
201                 System.out.print("New Summary: ");
202                 c.setSummary(sc.nextLine());
203 
204                 System.out.print("New MS Teams Link: ");
205                 c.setMsTeamsLink(sc.nextLine());
206 
207                 System.out.println("Course updated successfully!");
208                 c.displayCourse();
209             } else {
210                 System.out.println("Course not found.");
211             }
212         }
213 
```

```
219         /**
220          * Method: deleteCourse()
221          * Deletes a course after confirmation by the user.
222          */
223         private void deleteCourse() {
224             System.out.print("Enter Course Code to delete: ");
225             String code = sc.nextLine();
226             int index = findCourseIndex(code);
227 
228             //check if the course code entered exists or not.
229             if (index != -1) {
230                 Course c = courses.get(index);
231                 System.out.println("Course found:");
232                 c.displayCourse();
233 
234                 //confirmation to user to delete, must be (y/Y) to delete
235                 System.out.print("Confirm deletion? (Y/N): ");
236                 String confirm = sc.nextLine();
237                 if (confirm.equalsIgnoreCase("Y")) {
238                     courses.remove(index); // Remove from list
239                     System.out.println("Course deleted successfully!");
240                 } else {
241                     System.out.println("Deletion cancelled.");
242                 }
243             } else {
244                 System.out.println("Course not found.");
245             }
246         }
247 
248         /**
249          * Method: viewAllCourses()
250          * Displays all courses in the system.
251          */
252         private void viewAllCourses() {
253             /**check for all courses created and courses available in the database.
254             * if empty/none available, display the following message
255             * if courses exists, display all courses created.
256             */
257             if (courses.isEmpty()) {
258                 System.out.println("No courses available.");
259             } else {
260                 System.out.println("\n--- All Courses ---");
261                 for (Course c : courses) {
262                     c.displayCourse();
263                 }
264             }
265         }
266 
267         /**
268          * Method: findCourseIndex()
269          * Searches course list for a code. Returns index or -1 if not found.
270          */
271         private int findCourseIndex(String code) {
272             for (int i = 0; i < courses.size(); i++) {
273                 // equalsIgnoreCase: ignores uppercase/lowercase differences
274                 if (courses.get(i).getCourseCode().equalsIgnoreCase(code)) {
275                     return i; // Course found
```

```
280         /**
281          * Method: addStudent()
282          * Description: Adds a new student to the system
283          * - Input all details from user
284          * - Stores Student object in ArrayList
285          */
286         private void addStudent() {
287         // Ask for all inputs first
288         System.out.print("Enter Student ID: ");
289         String id = sc.nextLine();
290
291         System.out.print("Enter First Name: ");
292         String fname = sc.nextLine();
293
294         System.out.print("Enter Last Name: ");
295         String lname = sc.nextLine();
296
297         System.out.print("Enter Email: ");
298         String email = sc.nextLine();
299
300         System.out.print("Enter Phone: ");
301         String phone = sc.nextLine();
302
303         // Validate all inputs at once
304         if (id.isBlank() || fname.isBlank() || lname.isBlank() || email.isBlank() || phone.isBlank()) {
305             System.out.println("Student not added! All fields must be filled.");
306         } else {
307             // Only add if all fields are valid
308             students.add(new Student(id, fname, lname, email, phone));
309             System.out.println("Student added successfully!");
310         }
311     }
312
313         /**
314          * Method: searchStudent()
315          * Description: Search student by ID
316          * - Uses findStudentIndex()
317          * - Display the student's information if found
318          */
319         private void searchStudent() {
320             System.out.print("Enter Student ID to search: ");
321             String id = sc.nextLine();
322
323             int index = findStudentIndex(id); // search for student
324
325             if (index != -1) {
326                 System.out.println("Student found:");
327                 students.get(index).displayStudent(); // display student details
328             } else {
329                 System.out.println("Student not found."); // not found
330             }
331         }
```

```java
    /**
     * Method: editStudent()
     * - Updates student information by ID
     * - Prompts user to enter new data
     * - Ensures correct student is updated using index
     */
    private void editStudent() {
        System.out.print("Enter Student ID to edit: ");
        String id = sc.nextLine();

        int index = findStudentIndex(id);

        //if id entered exists in database, proceed to update student details
        if (index != -1) {
            Student s = students.get(index);

            System.out.print("New First Name: ");
            s.setFirstName(sc.nextLine());

            System.out.print("New Last Name: ");
            s.setLastName(sc.nextLine());

            System.out.print("New Email: ");
            s.setEmail(sc.nextLine());

            System.out.print("New Phone: ");
            s.setPhone(sc.nextLine());

            System.out.println("Student updated successfully.");
            s.displayStudent();
        } else {
            //if id entered does not exists, display the error message
            System.out.println("Student not found.");
        }
    }

```

```java
    /**
     * Method: deleteStudent()
     * - Find student index first
     * - Confirm deletion
     * - Remove from ArrayList
     * - Display all students after deletion
     */
    private void deleteStudent() {
        System.out.print("Enter Student ID to delete: ");
        String id = sc.nextLine();

        int index = findStudentIndex(id);

        //if id entered exists in database, proceed to update student details
        if (index != -1) {
            Student s = students.get(index);
            System.out.println("Student found:");
            s.displayStudent(); // display student's information before deletion

            //ask confirmation from user to delete, must be (y/Y)
            System.out.print("Confirm deletion? (Y/N): ");
            String confirm = sc.nextLine();

            if (confirm.equalsIgnoreCase("Y")) {
                students.remove(index); // delete student
                System.out.println("Student deleted successfully!");
                viewAllStudents(); // Show updated list
            } else {
                System.out.println("Deletion cancelled.");
            }
        } else {
            //if id entered does not exists, display the error message
            System.out.println("Student not found.");
        }
    }

     /**
     * Method: viewAllStudents()
     * - Loops through ArrayList and display each student
     * - Checks if list is empty first
     */
    private void viewAllStudents() {
        /**check for all students available in the database.
        * if empty/none available, display the following message
        * if students exists, display all students' information.
        */
        if (students.isEmpty()) {
            System.out.println("No students available.");
        } else {
            System.out.println("\n--- All Students ---");
            for (Student s : students) {
                s.displayStudent();
            }
        }
    }
```

```
424         /**
425          * Method: findStudentIndex()
426          * - Returns index of student in ArrayList
427          * - Linear search
428          * - Returns -1 if not found
429          * - equalsIgnoreCase() to ignore case sensitivity in ID
430          */
431         private int findStudentIndex(String id) {
432             for (int i = 0; i < students.size(); i++) {
433                 // equalsIgnoreCase: allows case-insensitive comparison
434                 if (students.get(i).getStudentID().equalsIgnoreCase(id)) {
435                     return i; // student found, display student's information
436                 }
437             }
438             // -1 indicates student not found
439             return -1;
440         }
441     }
```

14.0 REFERENCES

Moodle Official Website. https://moodle.org/

Moodle Codebase Repository. https://github.com/moodle/moodle